

HTML to Source Code Converter

1. Objective

Develop a C program that converts C source code into an HTML file with syntax highlighting for improved readability in web browsers.

2. Description

The application reads a C source file, identifies tokens such as keywords, comments, strings, header files, numeric constants and preprocessor directives, and generates an equivalent HTML file using CSS classes for syntax highlighting.

3. Features

- Converts C source code to HTML
- Highlights keywords
- Highlights comments
- Highlights string and character literals
- Highlights header files
- Highlights numeric constants
- Highlights preprocessor directives
- Uses CSS for styling
- Generates browser-ready HTML output

4. Technologies Used

Language: C

Compiler: GCC

Platform: Linux (Ubuntu)

Editor: VS Code

Styling: HTML & CSS

5. Project Structure

s2html_main.c

s2html_conv.c

s2html_conv.h

s2html_event.c

s2html_event.h

styles.css

Makefile

test.c

6. Algorithm

1. Read the input C source file.
2. Parse each character.
3. Detect tokens based on parser state.
4. Wrap detected tokens in HTML `<span>` tags.
5. Write formatted output to an HTML file.
6. Close the HTML document.

7. Compilation

```
gcc s2html_main.c s2html_conv.c s2html_event.c -o s2html
```

8. Execution

```
./s2html test.c
```

9. Sample Output

An HTML file (test.c.html) is generated and can be opened in a web browser to view syntax-highlighted source code.

10. Applications

Educational tools, source code visualization, documentation generation, and code sharing.

11. Conclusion

The project successfully converts C source code into a syntax-highlighted HTML document, making code easier to read and present.